Jian Feng

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Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor's degree, Computer Science, Finance, and Statistics *Waterloo, ON, Canada*

Skills

Languages:	Python, C/C++, Java, C#, Go, Bash, Typescript, Javascript, SQL, R
AI/ML:	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, scikit-learn, Microsoft Agent Framework, OpenCV, pandas
Frameworks:	.Net, Spring Boot, React, Angular, Node.js, Flask, Rails, FastAPI
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps:	AWS, GCP, Azure, Docker, K8s, Terraform, GitHub Actions, Linux, Jenkins
Tools:	Git, Jira, Confluence

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Architected a reusable pre-validation and skip-on-failure execution pattern across enterprise ServiceNow catalog task workflows; eliminated manual exception handling overhead and enabled resilient, non-blocking asynchronous orchestration for global 1k+ daily cross-system operations.
- Migrated the SNOW orchestrator core from legacy basic authentication to secure GitHub App token exchange via PyGithub; established strict mfc-ets team-level validation, automated permission scopes, and end-to-end audit-trail compliance across catalog automations.
- Enhanced the GOCC orchestrator with live Kubernetes cluster command integration for real-time AKS cert-SAN verification; empowered the Foundry LLM agent to automatically cross-examine GitOps pull requests against live cluster states, successfully catching misconfigurations prior to production deployment.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd. Guangdong, China
Software Engineer *Jun 2024 – Nov 2024*

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

Projects

MiniKV | Go

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

GhostPilot

- Invisible Windows overlay streams WASAPI loopback audio through partial/final Azure ASR into a text LLM.
- Alt+P vision path: Qt area capture ⟶ JPEG ⟶ vision LLM, result folded back into the text-LLM prompt.
- Local RAG over `knowledge/` injects user-curated context so answers stay grounded across long sessions.

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

Anchor | Python

- Splunk drift agent fingerprints a healthy window, then ranks diffs against later windows with LLM-narrated SPL suggestions.
- Feedback loop reweights signals and surfaces recurring unresolved drifts as blind spots.

CabStream ETL | Python

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging ⟶ dims/facts ⟶ marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

Achievements

- Scaled UX Research Hub 32× (10⟶318 members) as Director; negotiated Google UXR keynote at $0 cost and brokered TikTok chapter cross-promotion for sustained recruitment growth.
- Velocity Co-op Problem Award (top submission among co-op cohort); 3rd Place, SAF Investment Research Challenge; Finalist, SAF Strategy Case Competition, Moonshot Hackathon Winner

Coursework

IBM Vector Database, Stanford Machine Learning, UCI Database Design & Operational BI, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning